QP Code:- 25900894

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Course (DSC) - MG1DSCMAT100 – GROUND ROOTS OF MATHEMATICS

(2024 ADMISSION ONWARDS)

SCHEME FOR VALUATION

Duration: 2 Hours **Maximum Marks: 70**

Part A

Short Answer Type Questions

Answer any **5** Questions. Each Question carries **2** marks.

1. Predicate is "**spends more than five hours every weekday in class".** …2 Marks **[U] / [CO1]**
2. True … 2 Marks **[U] / [CO1]**
3. Let $y = \sqrt{x+5}$ … $\frac{1}{2}$ Marks **[A] / [CO3]**

   For getting, $y^2 = x + 5$ … 1 Mark

   For writing, $x = y^2 - 5$ … 1 Mark

   Getting, $f^{-1}(x) = y = x^2 - 5$ … $\frac{1}{2}$ Marks

   Total marks should not exceed 2.
4. For correct definition … 2 Marks **[K] / [CO2]**
5. For writing product rule … 1 Mark **[A] / [CO4]**

   Applying product rule and getting, $\frac{dy}{dx} = x\cos x + \sin x$ … 2 Marks

   Total marks should not exceed 2.
6. Differentiating w.r.t. x and getting, $\cos y + x\,(-\sin y)\frac{dy}{dx} = \frac{dy}{dx}$ 2 Marks **[A] / [CO4]**

   For writing $\cos y = \frac{dy}{dx}(1 + x\sin y)$ … 1 Mark

   For obtaining, $\frac{dy}{dx} = \frac{\cos y}{1+x\sin y}$ … 1 Mark

   Total marks should not exceed 2.
7. $f'(x) = 3x^2 - 6x$ … 1 Mark **[U] / [CO5]**

   $f''(x) = 6x - 6$ … $\frac{1}{2}$ Marks

   $f''(x) = 0 \Rightarrow x = 1$ … $\frac{1}{2}$ Marks

   For writing f is concave up on $(1, \infty)$ and f is concave down on $(-\infty, 1)$. … 1 Mark

Inflection point is $(1, -1)$. … 1 Mark

Total marks should not exceed 2.

8. For writing the limit has the indeterminate form $\frac{\infty}{\infty}$ … 1 Mark **[A] / [CO6]**

   Applying L'Hôpital's rule and writing $\lim_{x\to\infty} \frac{\ln x}{2\sqrt{x}} = \lim_{x\to\infty} \frac{\frac{1}{x}}{2\times\frac{1}{2\sqrt{x}}}$ … 1 Mark

   $= \lim_{x\to\infty} \frac{\sqrt{x}}{x} = \lim_{x\to\infty} \frac{1}{\sqrt{x}} = 0$ … 1 Mark

   Total marks should not exceed 2.

Part B

Short Essay Type Questions

Answer any **5** Questions. Each Question carries **6** marks.

9. (a)$p \wedge q$ … 2 Marks **[U] / [CO1]**

   (b) $\neg p \wedge \neg q$ … 2 Marks

   (c) $p \to q$ … 2 Marks

10. For writing, not logical equivalent … 2 Marks **[U] / [CO1]**

    For counter example … 4 marks

11. $y = x^2 + 1$ … 3 Marks **[U] / [CO2]**

    $y = x^2 - 2$ … 3 Marks

12. For correct definition of one-to-one function … 2 Marks **[A] / [CO3]**

    Horizontal line test … 2 Marks

    For drawing the graph of $y = \cos x$ … 2 Marks

    For applying horizontal line test and concluding that $y = \cos x$ is not one-to-one … 2 Marks

    Total marks should not exceed 6.

13. $F'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$ … 3 Marks **[A] / [CO4]**

    $F'(1) = \frac{g(1)f'(1) - f(1)g'(1)}{(g(1))^2}$ … 2 Marks

    $F'(1) = \frac{3\times 2 - (-1)\times(-1)}{(3)^2} = \frac{5}{9}$ … 2 Marks

    Total marks should not exceed 6.

14. For applying quotient rule and getting $\frac{dx}{dt} = \frac{3t\times 2t - (t^2+1)\times 3}{(3t)^2}$ … 4 Marks **[A] / [CO4]**

    $= \frac{6t^2 - 3(t^2+1)}{9t^2}$ … 1 Mark

    $= \frac{3t^2 - 3}{9t^2}$ … 1 Mark

    $= \frac{t^2 - 1}{3t^2}$ … 1 Mark

Total marks should not exceed 6.

15. $f'(x) = 3 \times \frac{5}{3}x^{\frac{5}{3}-1} - 15 \times \frac{2}{3}x^{\frac{2}{3}-1}$ … 3 Marks **[U] / [CO5]**

$= 5x^{\frac{2}{3}} - 10x^{\frac{-1}{3}}$ … 1 Mark

$= 5x^{-\frac{1}{3}}(x-2)$ … 1 Mark

$= \frac{5(x-2)}{x^{\frac{1}{3}}}$ … 1 Mark

For writing definition of critical points … 1 Mark

Critical points are $x = 2$ and $x = 0$. … 2 Marks

Total marks should not exceed 6.

16. 1^{∞} form … 2 Marks **[A] / [CO6]**

Let $L = \lim_{x\to 0}(1+2x)^{-\frac{3}{x}}$ … 1 Mark

$\ln L = \ln \lim_{x\to 0}(1+2x)^{-\frac{3}{x}}$ … 1 Mark

$\ln L = \lim_{x\to 0} \ln (1+2x)^{-\frac{3}{x}}$ … 1 Mark

$\ln L = \lim_{x\to 0} -\frac{3}{x} \ln (1+2x) \quad \left(\frac{0}{0}\ form\right)$ … 1 Mark

Apply L'Hôpital's rule and getting, $\ln L = \lim_{x\to 0} \frac{-3\times\frac{1}{1+2x}\times 2}{1}$ … 2 Marks

$= \lim_{x\to 0} \frac{-6}{1+2x} = -6$ … 1 Mark

$L = e^{-6}$. … 1 Mark

Total marks should not exceed 6.

Part C

Essay Type Questions

Answer any **3** Questions. Each Question carries **10** marks.

17. (a)$(p \wedge q) \to (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q)$ … 2 Marks **[U] / [CO1]**

$\equiv (\neg p \vee \neg q) \vee (p \vee q)$ [by the first De Morgan law] … 1 Mark

$\equiv \neg p \vee (\neg q \vee \mathrm{p}) \vee q)$ [by the associative law] … 1 Mark

$\equiv \neg p \vee (\mathrm{p} \vee \neg q) \vee q)$ [by the commutative law] … 1 Mark

$\equiv (\neg p \vee p) \vee (\neg q \vee q)$ [by the associative law] … 1 Mark

$\equiv \mathbf{T} \vee \mathbf{T} \equiv \mathbf{T}$ … 1 Mark

OR

For the truth table … 6 Marks

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

$(p \wedge q) \rightarrow (p \vee q)$ is always true. So, it is a tautology. … 1 Mark

(b) $\neg(p \rightarrow q) \equiv \neg(\neg p \vee q)$ [since $p \rightarrow q \equiv \neg p \vee q$] … 2 Marks

$\equiv \neg(\neg p) \wedge \neg q$ [by the second De Morgan law] … 2 Marks

$\equiv p \wedge \neg q$ [by the double negation law] … 1 Mark

OR

For the truth table … 4 Marks

p	q	$\neg$q	p $\rightarrow$ q	$\neg$(p $\rightarrow$ q)	p $\wedge\neg q$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

Since the truth values of $\neg(p \rightarrow q)$ and $p \wedge \neg q$ agree, $\neg(p \rightarrow q) \equiv p \wedge \neg q$. … 1 Mark

Total marks should not exceed 10.

18. Definition of even function … 2 Marks **[U] / [CO2]**

Definition of odd function … 2 Marks

For proving $f(x)$ is even. … 3 marks

For proving $f(x)$ is odd. … 3 marks

19. (a) $\frac{f}{g} = \frac{\sqrt{x}}{\sqrt{1-x}} = \sqrt{\frac{x}{1-x}}$ … 2 Marks **[U] / [CO2]**

Domain of $f = [0, \infty)$ and domain of $g = (-\infty, 1]$ … 2 Marks

Domain of $\frac{f}{g}$ is $[0,1)$. … 2 Marks

$\frac{g}{f} = \frac{\sqrt{1-x}}{\sqrt{x}} = \sqrt{\frac{1-x}{x}}$ … 2 Marks

Domain of $\frac{g}{f}$ is $(0,1]$. … 2 Marks

(b) For replacing x by $x + 1$ in the given equation (shifting 1 unit to the left) … 2 Marks

For shifting 1 unit down … 2 Marks

For getting the required equation as $y = (x+1)^3 - 1$ … 1 Mark

Total marks should not exceed 10.

20. (a) $\frac{dy}{dx} = 6(3x^2 + 2x - 1)^5(6x + 2)$ … 5 Marks **[A] / [CO4]**

    (b) $\frac{dy}{dx} = 4(x^2 \tan x)^3(x^2 sec^2 x + 2x \tan x)$ … 5 Marks

21. (a) For getting f is increasing on $[0, \infty)$. … 5 Marks **[U] / [CO5]**

    (b) For getting concave upon $\left(-\infty, \frac{4}{3}\right) \cup (4, \infty)$ … 3 Marks

    (c) For getting the function is concave down on $\left(\frac{4}{3}, 4\right)$. … 3 Marks

Total marks should not exceed 10.

22. (a) The given limit is an indeterminate form of type $\frac{0}{0}$. … 2 Marks **[A] / [CO6]**

    Applying L'Hôpital's rule to get $\lim_{t\to 0} \frac{1-e^t}{te^t} = \lim_{t\to 0} \frac{-e^t}{te^t+e^t}$ … 3 Marks

    $= \lim_{t\to 0} \frac{-e^t}{e^t(t+1)} = \lim_{t\to 0} \frac{-1}{t+1} = -1$ … 2 Marks

    (b) The given limit is an indeterminate form of type $\frac{0}{0}$. … 2 Marks

    Applying L'Hôpital's rule to get $\lim_{x\to+\infty} \frac{x^{-\frac{4}{3}}}{\sin\left(\frac{1}{x}\right)} = \lim_{x\to+\infty} \frac{-\frac{4}{3}x^{-\frac{7}{3}}}{-\frac{1}{x^2}\cos\left(\frac{1}{x}\right)}$ … 3 Marks

    $= \lim_{x\to+\infty} \frac{\frac{4}{3}x^{-\frac{1}{3}}}{\cos\left(\frac{1}{x}\right)} = \frac{0}{1} = 0$ … 2 Marks

Total marks should not exceed 10.

N.B: Suitable stage marks may be fixed by the examiner for alternate correct answers.